Human Review: Zero Thickness and Hilbert Bi-Lipschitz Embeddings

Original automatic proof: `solution_packet.pdf`
Checked date: 18 June 2026
Status: Manually checked.
Verdict: The proof appears correct.

Human Comments

The proposed argument appears to give a correct negative answer to the second open question in Margaris–Robinson, *Embedding properties of sets with finite box-counting dimension*. The question asks whether thickness exponent zero forces the existence of a nonlinear bi-Lipschitz embedding into either a Euclidean space or a Hilbert space. The construction shows that this is not so.

The idea is to hide larger and larger Hamming cubes at smaller and smaller scales. For each m, the Hamming cube

$$Q_m = \{0,1\}^m$$

is embedded isometrically into $E_m = \ell_\infty(Q_m)$ by the pointed Kuratowski embedding

$$\iota_m(x) = \big(d_m(x,y) - d_m(0_m,y)\big)_{y\in Q_m}.$$

This is indeed isometric: for $x, x' \in Q_m$,

$$\|\iota_m(x) - \iota_m(x')\|_\infty = \sup_{y\in Q_m} |d_m(x,y) - d_m(x',y)| = d_m(x,x').$$

The upper bound follows from the triangle inequality, while equality is obtained by taking $y = x$.

These cubes are then placed in the c_0-sum

$$B = \left(\bigoplus_{m=1}^{\infty} E_m\right)_{c_0}$$

after scaling the m-th copy by

$$s_m = \frac{R_m}{m}, \qquad R_m = 2^{-m^3}.$$

Thus the m-th level has diameter at most R_m, and the compact set

$$X = \{0\} \cup \{u_{m,x} : m \in \mathbb{N},\ x \in Q_m\}$$

accumulates only at zero.

The box-counting estimate is correct. If

$$R_{M+1} < \varepsilon \le R_M,$$

then all levels with $m > M$ are contained in the ε-ball around zero, while the first M levels can be covered pointwise. Hence

$$N(X,\varepsilon) \le 1 + \sum_{m=1}^{M} 2^m \le 2^{M+1}.$$

Since $-\log R_M = M^3 \log 2$, it follows that

$$\overline{\dim}_B(X) = 0.$$

The thickness exponent is bounded above by the upper box-counting dimension: an ε-cover with centers $x_1, \ldots, x_N$ gives an ε-approximating subspace $\operatorname{span}\{x_1, \ldots, x_N\}$. Therefore

$$\tau(X, B) = 0.$$

The obstruction to Hilbert bi-Lipschitz embeddability is also correct. If $F : X \to H$ were bi-Lipschitz with constants $0 < a \leq b$, then restricting F to the m-th level would give a Hilbert embedding of Q_m with distortion at most b/a, uniformly in m, because the level is just a scaled isometric copy of Q_m.

This contradicts the classical Enflo cube estimate: every embedding of the m-dimensional Hamming cube into a Hilbert space has distortion at least $\sqrt{m}$. One way to see this is through the Walsh expansion on $\{-1, 1\}^m$, which gives the cube Poincare inequality

$$\mathbb{E}_\varepsilon \|G(\varepsilon) - G(-\varepsilon)\|^2 \leq \sum_{j=1}^{m} \mathbb{E}_\varepsilon \|G(\varepsilon) - G(\varepsilon^{(j)})\|^2.$$

If G has Lipschitz constant at most β, the right-hand side is at most $m\beta^2$, so some antipodal pair has image distance at most $\beta\sqrt{m}$. The lower Lipschitz bound gives distance at least αm for the same pair. Hence $\beta/\alpha \geq \sqrt{m}$. This is incompatible with one fixed distortion constant for all levels.

Literature Check

A preliminary search did not reveal an existing solution of the stated open question. The citation trail from the published Margaris–Robinson article appears very small: the relevant later work found is Margaris's paper on almost bi-Lipschitz embeddings using covers of balls centred at the origin, which proves an almost bi-Lipschitz result under an almost-homogeneity condition, not a genuine bi-Lipschitz embedding theorem for thickness zero. Targeted searches for the exact phrasing of the question and for variants involving "thickness exponent", "bi-Lipschitz", "Hilbert", and "Euclidean" did not turn up a prior answer.

The main obstruction used here, however, is standard: it is the classical Enflo/Hamming-cube lower bound for Hilbert distortion. Thus the result should be presented as an apparently correct and natural counterexample built from a standard metric-geometry obstruction, pending author feedback or a fuller literature check.

Review Outcome

Archive this packet as an apparently correct counterexample to the second open question of Margaris–Robinson. The proof is short, conceptually clear, and the central mechanism is sound: thickness and box-counting dimension miss the high-dimensional Hamming-cube geometry hidden at rapidly shrinking scales, whereas any Hilbert bi-Lipschitz embedding would have to preserve it with one uniform distortion constant.